# Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

## Position

| | |
|---|---|
| **Meaning in my words**<br>How would you explain position to a friend? | **How can you tell?**<br>The doorway is zero and east is positive. Where is a cart at −3 m? |
| **Example or sketch**<br>How would you mark a cart 2 m east of the doorway on a number line? | **Non-example and why**<br>"I walked 7 m." Do we know your position now? |

## Distance

| | |
|---|---|
| **Meaning in my words**<br>How would you explain distance to a friend? | **How can you tell?**<br>A cart goes 3 m right and 2 m left. How much ground do its wheels cover? |
| **Example or sketch**<br>Draw a trip that covers 6 m and finishes at its starting point. | **Non-example and why**<br>A partner writes −5 m for distance. What does the minus sign tell you? |

# Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

## Displacement

| | |
|---|---|
| **Meaning in my words**<br>How would you explain displacement to a friend? | **How can you tell?**<br>Which two marks tell you the displacement even if the cart turns around? |
| **Example or sketch**<br>Go from 2 m to 6 m and back to 4 m. What single arrow shows the displacement? | **Non-example and why**<br>Someone adds both legs and gets 6 m. What did they find? |

## Scalar / vector

| | |
|---|---|
| **Meaning in my words**<br>How would you explain the difference between a scalar and a vector? | **How can you tell?**<br>Which tells you which way to draw an arrow: 5 m or 5 m west? |
| **Example or sketch**<br>Give a scalar and a vector for a 4 m walk straight east. | **Non-example and why**<br>Two students each walk 4 m, one east and one west. Are their displacements equal? |

# Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

## Put start, turn, and finish on a number line. Label distance on the path and displacement on the shortcut.

Choose zero, direction, and scale.



**My marks mean**

# On assignment: Displacement

SECTION E / BACK / FINAL MEETING ONLY Name _____

**A survey rover tracks XJ's model from +2 to +8 to +5 km. Dispatch needs both mileage and final location.**

## PICK A JOB / DELIVER ONE PRODUCT

☐ **Route engineer**

Draw the two legs for the mileage log and a single arrow for the return crew. Give dispatch the two different numbers.

☐ **Radio dispatcher**

Guide a new driver to the finish. Give the trip length, then its change in position, and tell the driver which number locates the finish from the start.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

## INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.